Cambridge (CIE) A Level Physics

Your notes

Temperature Scales

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Measurement of temperature

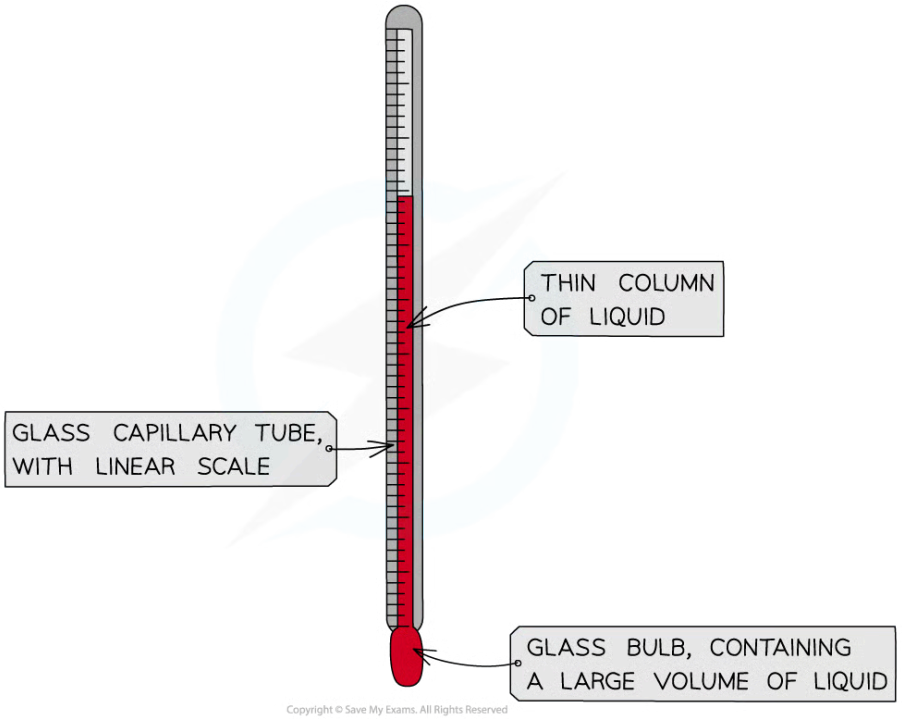
- **Temperature** is measured using a **physical property** that **varies with temperature**
  - A **thermometer** is any device used to **measure temperature**
- Each type of thermometer uses the physical property of a material that varies with temperature, for example:
  - the density of a liquid
  - the volume of a gas at constant pressure
  - the resistance of a metal
  - the e.m.f. of a thermocouple
- Each type of thermometer is calibrated at two or more known temperatures (commonly the boiling and melting points of water, 0 °C and 100 °C respectively) and the scale is divided into equal divisions

The density of a liquid

- A liquid-in-glass thermometer depends on the **density change of a liquid** (commonly mercury)
  - The thin glass capillary tube contains a liquid that **expands** with temperature
  - The scale on the side of the tube is measured based on the **length of liquid** within the tube

A liquid-in-glass thermometer

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As the bulb is heated, the liquid expands and moves along the capillary tube

Volume of a gas at constant pressure

- The **volume** of an ideal gas is **directly proportional** to its **temperature** when at constant pressure (Charles's law)

$$V \propto T$$

- As the **temperature** of the gas **increases**, its **volume increases** and vice versa
- A gas thermometer must be calibrated when the temperature of a gas at a certain volume is known
  - A temperature scale can be determined depending on how quickly the gas expands with temperature

Resistance of a metal

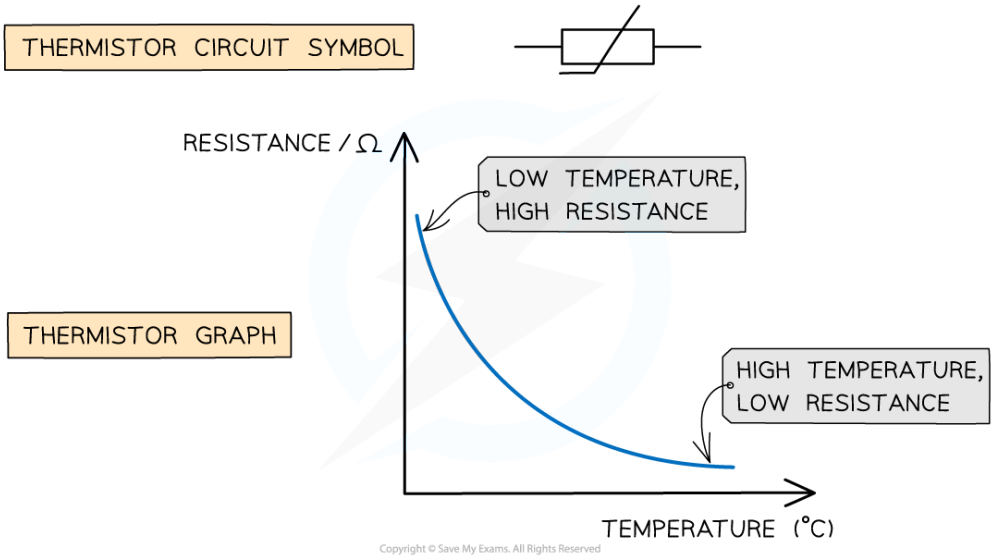
- Recall that **electrical resistance changes with temperature** e.g. the resistance of a filament lamp increases when the current passing through it also increases
  - For metals: resistance increases with temperature at a steady rate
  - For thermistors: resistance changes rapidly over a narrow range of temperatures
- As a **thermistor** gets **hotter**, its **resistance decreases**
  - A thermistor can be used to measure a range of temperatures
- The relationship between the resistance and temperature of a thermistor is **non-linear**

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- The graph of temperature against resistance is a **curved line**, so the thermistor has to be calibrated

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Graph of resistance against temperature

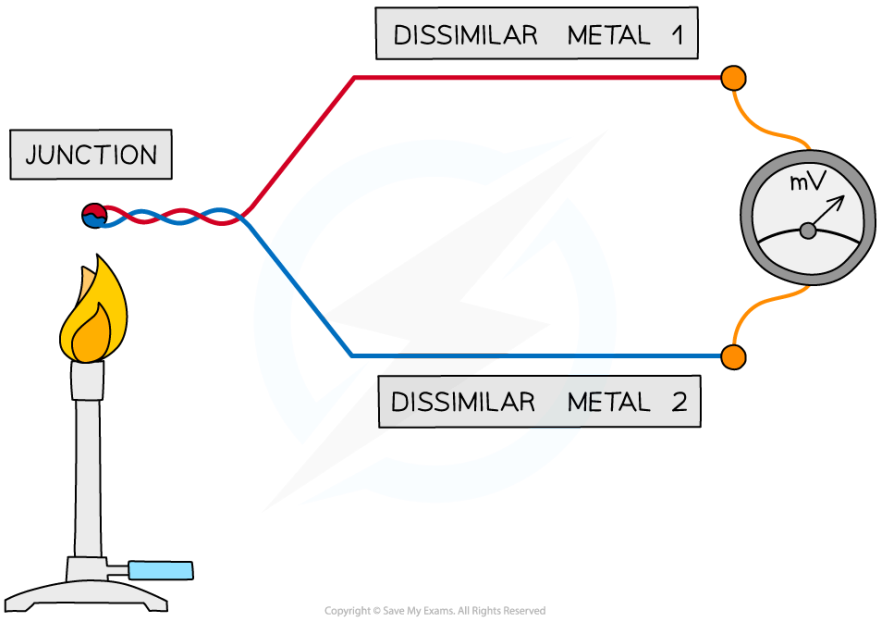


As the temperature of a thermistor increases, its resistance decreases.

E.m.f. of a thermocouple

- A **thermocouple** is an electrical device used as the **sensor** of a thermometer
- A thermocouple has **two wires** of different, or dissimilar, metals attached to each other to create two **junctions**
  - The other ends of the wires are connected to a voltmeter
- When there is a **temperature difference** between the junctions, an **e.m.f.** is **produced** between the two free ends of the wires and measured by the voltmeter
  - The greater the difference in temperature between the wires, the greater the e.m.f

Diagram showing a thermocouple

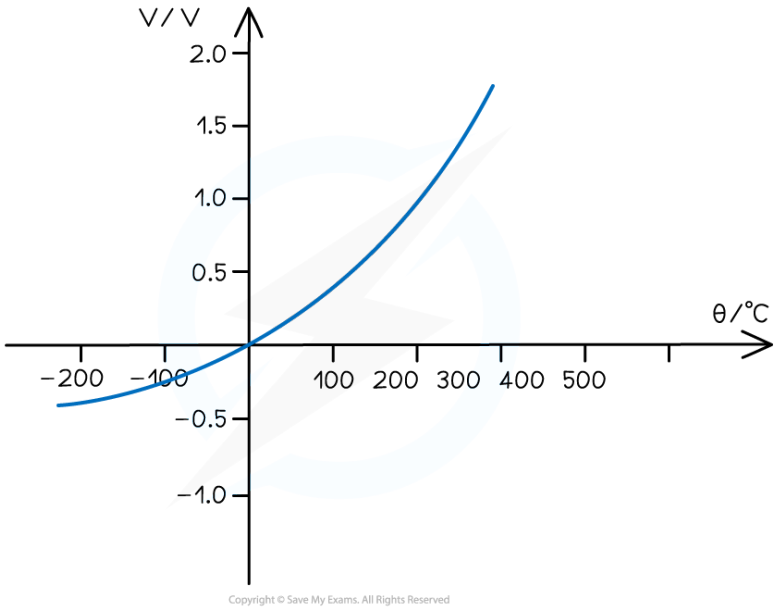


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A thermocouple consists of two dissimilar wires connected together

- A thermocouple requires calibration since the e.m.f. does not vary linearly with temperature
- The graph against e.m.f. and temperature is a positive, curved line

Graph showing e.m.f against temperature



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The e.m.f. and temperature are not directly proportional in a thermocouple.

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Examiner Tips and Tricks

Remember to relate how the temperature is measured for different types of thermometer back to the scenario in the question. For example, make sure you say: the temperature increases as the **volume of gas increases** or the temperature increases as the **e.m.f. between the two wires increases**.



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Scale of thermodynamic temperature

- Temperature is a **scalar** quantity measured in **degrees Celsius** (**°C**) or **kelvin** (**K**)
  - Kelvin is the **SI base unit** for temperature

The Kelvin scale

- The **scale of thermodynamic temperature** does **not depend** on the **property** of any particular substance
  - The **thermodynamic temperature scale** is also known as the **Kelvin scale**
- The thermodynamic scale is an **absolute scale** because it is not defined in terms of the properties of any particular substances
  - This is unlike the **degrees Celsius (°C)** temperature scale which is based on the **properties of water**

The degrees Celsius scale

- When the **degrees Celsius (°C)** temperature scale was created
  - The freezing point of water was taken as 0 °C
  - The boiling point was taken as 100 °C
- The Celsius scale measures the temperature in a **liquid-in-glass thermometer**
  - The expansion of the liquid is not always linear

Using the kelvin scale

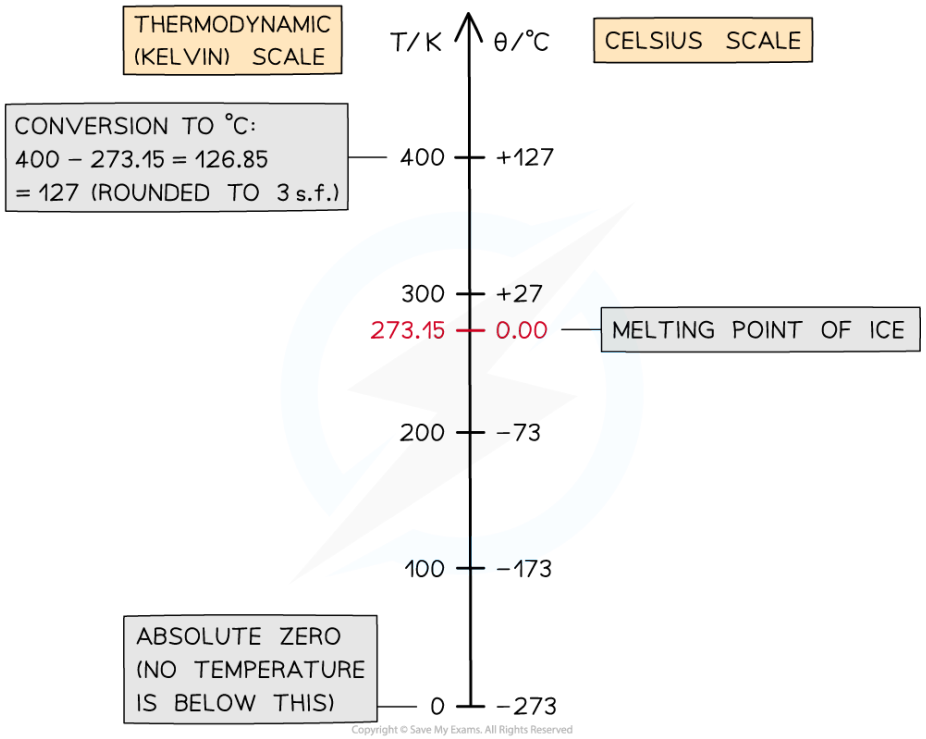
- Temperatures in the Celsius scale, θ, and temperatures in the Kelvin scale, T, can be converted using the equation:

$$\theta / °C = T / K - 273.15$$

$$T / K = \theta / °C + 273.15$$

Conversion chart for Kelvin and Celsius

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Conversion chart relating the temperature on the Kelvin and Celsius scales.

- The **divisions** on both scales are **equal** which means that:

A change in a temperature of 1 K is equal to a change in temperature of 1 °C

Worked Example

In many ideal gas problems, room temperature is considered to be 300 K.

What is this temperature in Celsius?

Answer

Step 1: Kelvin to Celsius equation

$$\theta / ^\circ C = T / K - 273.15$$

Step 2: Substitute in value of 300 K

$$300\ K - 273.15 = 26.85\ ^\circ C$$

Examiner Tips and Tricks

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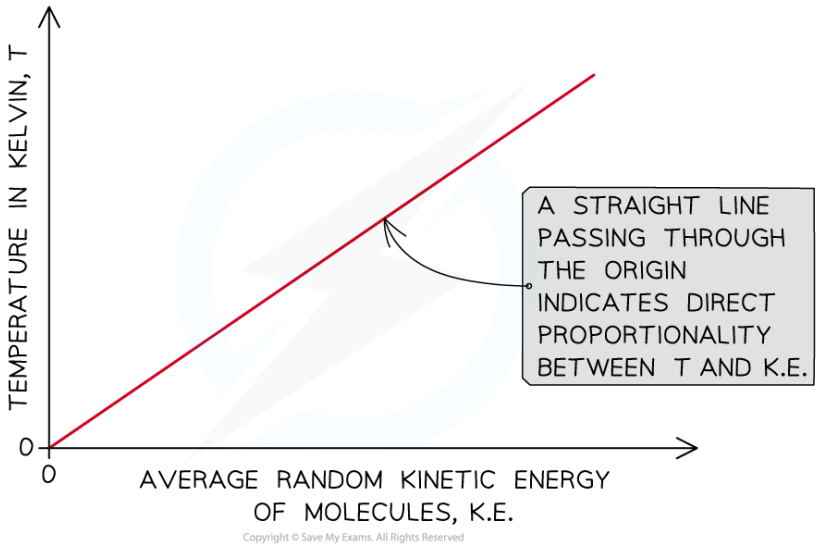
If you forget in the exam whether it's +273.15 or –273.15, just remember that 0 °C = 273.15 K. This way, when you know that you need to +273.15 to a temperature in degrees to get a temperature in Kelvin. For example: 0 °C + 273.15 = 273.15 K.

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Absolute zero

- The **lowest possible temperature** on the thermodynamic temperature scale, known as **absolute zero**, is **zero Kelvin**
  - It is not possible to have a temperature lower than 0 K
  - So a temperature in Kelvin will **never** be a negative value
- **Absolute zero** corresponds to the temperature at which the **average kinetic energy of the molecules** is at its **minimum**
  - It is not possible to remove any more energy from a system with a temperature of 0 K
  - Even in space, the temperature is roughly 2.7 K, just above absolute zero
- The **absolute temperature** of a body is **directly proportional to the average kinetic energy of the molecules** within the body

Temperature-kinetic energy graph



At absolute zero, the temperature is zero Kelvin and the kinetic energy of the molecules is zero